

Notations

1. Greek Alphabet

Uppercase	Lowercase	Name
A	α	Alpha
B	β	Beta
Γ	γ	Gamma
Δ	δ	Delta
E	ε	Epsilon
Z	ζ	Zeta
H	η	Eta
Θ	θ	Theta
I	ι	Iota
K	κ	Kappa
Λ	λ	Lambda
M	μ	Mu
N	ν	Nu
Ξ	ξ	Xi
O	$\omicron$	Omicron
Π	π	Pi
P	ρ	Rho
Σ	σ	Sigma
T	τ	Tau
Y	υ	Upsilon
Φ	φ	Phi
X	χ	Chi
Ψ	ψ	Psi
Ω	ω	Omega

2. Logical & Set Notation

Symbol	Name	Read As	Example
$\neg$	Negation	Not	$\neg P$ (Not P)
$\wedge$	Conjunction	And	$P \wedge Q$ (P and Q)
$\vee$	Disjunction	Or	$P \vee Q$ (P or Q)
$\oplus$	Exclusive Or	Either... or	$P \oplus Q$ (Either P or Q , but not both)
$\rightarrow$	Implication	If... then / Implies	$P \rightarrow Q$ (If P , then Q)
$\leftrightarrow$	Biconditional	If and only if	$P \leftrightarrow Q$ (P if and only if Q)
$\forall$	Universal Quantifier	For all / For every	$\forall x P(x)$ (For all x , $P(x)$ holds)
$\exists$	Existential Quantifier	There exists / For some	$\exists x P(x)$ (There exists an x such that $P(x)$)
$\nexists$	Non-existential Quantifier	There does not exist	$\nexists x P(x)$ (No x satisfies $P(x)$)
$\exists!$	Uniqueness Quantifier	There exists a unique	$\exists! x P(x)$ (There is exactly one x where $P(x)$ holds)
$\top$	Tautology	Top / Always true	$P \vee \neg P \equiv \top$
$\perp$	Contradiction	Bottom / Always false	$P \wedge \neg P \equiv \perp$
$\vdash$	Syntactic Turnstile	Proves / Yields	$\Gamma \vdash P$ (Γ proves P)
$\nvdash$	Non-provability	Does not prove	$\Gamma \nvdash P$ (Γ cannot prove P)
$\models$	Semantic Turnstile	Entails / Satisfies	$\Gamma \models P$ (Γ semantically entails P)

Symbol	Name	Read As	Example
$\not\models$	Non-entailment	Does not entail	$\Gamma \not\models P$ (Γ does not entail P)
$\equiv$	Logical Equivalence	Is logically equivalent to	$P \rightarrow Q \equiv \neg P \vee Q$
$\in$	Element of	Belongs to	$x \in A$ (x belongs to set A)
$\notin$	Not Element of	Does not belong to	$x \notin A$ (x is not in set A)
$\subseteq$	Subset	Is a subset of	$A \subseteq B$ (A is contained in or equal to B)
$\subset$	Strict Subset	Is a proper subset of	$A \subset B$ (A is strictly contained in B)
$\not\subseteq$	Not a Subset	Is not a subset of	$A \not\subseteq B$ (A is not a subset of B)
$\emptyset$	Empty Set	Empty set / Null set	$A = \emptyset$ (Set A has no elements)
$\cup$	Union	Union of	$A \cup B$ (Elements in A or B or both)
$\cap$	Intersection	Intersection of	$A \cap B$ (Elements in both A and B)
$\setminus$	Set Difference	Minus / Relative complement	$A \setminus B$ (Elements in A but not in B)

e.g.

$$\forall x(x \in A \cup B \rightarrow (x \in A \vee x \in B)) \wedge \nexists y(y \in \emptyset \wedge y \notin (A \setminus B))$$

Read as: “For all x , x being an element of A union B implies that x is in A or x is in B ; and there exists no y such that y is in the empty set and y is not in A minus B .”

e.g.

$$(\Gamma \models P \oplus Q) \equiv (\Gamma \vdash P \vee Q) \wedge (\Gamma \not\vdash P \wedge Q) \wedge (P \wedge \neg P \equiv \perp)$$

Read as: “Gamma semantically entails P exclusive-or Q is logically equivalent to saying: Gamma proves P or Q, Gamma does not prove P and Q, and P and not P is a contradiction.”

e.g.

$$(\forall \alpha \in A, \exists! \beta \in B : f(\alpha) = \beta) \rightarrow (A \subseteq B \wedge A \not\subseteq \emptyset)$$

Read as: “If for every alpha in A there exists a unique beta in B such that f of alpha equals beta, then A is a subset of B and A is not a subset of the empty set.”

3. Mathematical Terms

- Definition: An explicit statement of the precise meaning of a mathematical term, symbol or proof.
- Theorem: A mathematical statement that has been proven to be true.
- Axiom / Postulate: A self evident truth or statement accepted as true without proof.
- Proposition: A standalone mathematical statement that is proven using axioms.
- Lemma: A minor or subsidiary theorem used as stepping stone to prove a larger theorem.
- Corollary: A proposition or statement that follows directly and easily from a previously proven theorem.
- Hypothesis: A proposed explanation or conditional assumption used as a basis of further reasoning or proof.
- Conjecture: A statement or mathematical hypothesis that is suspected to be true based on preliminary evidence or intuition but has not yet been formally proved or disproved.